

Paper 1: Computer systems

Chapter 1: Data representation:

1.1 Number systems:

Question 1:

Binary is a number system used by computers.

(a) Tick (✓) **one** box to show which statement about the binary number system is correct.

- | | |
|---------------------------------|--------------------------|
| A It is a base 1 system | ☐ |
| B It is a base 2 system | ☐ |
| C It is a base 10 system | ☐ |
| D It is a base 16 system | ☐ |

[1]

(b) Denary numbers are converted to binary numbers to be processed by a computer.

Convert these **three** denary numbers to 8-bit binary numbers.

50

102

221

[3]

Working space

.....

.....

.....

(c) Binary numbers are stored in registers.

Negative denary numbers can be represented as binary using two's complement.

Complete the binary register for the denary number -78

You must show all your working.

Working space

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.....

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.....

.....

Register:

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[2]

(d) Two 8-bit binary numbers are given.

Add the **two** 8-bit binary numbers using binary addition.

Give your answer in binary. Show all your working.

$$\begin{array}{r} 00110011 \\ + 01100001 \\ \hline \end{array}$$

[3]

(e) Two binary numbers are added by a computer and an overflow error occurs.

Explain why the overflow error occurred.

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.....

.....

..... [2]

Question 2:

Binary numbers can be converted to hexadecimal.

- (a) Convert the **two** binary numbers to hexadecimal.

10010011

00001101 [4]

Working space

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.....

- (b) A value is stored as a binary number in a register.

0	1	1	1	1	0	1	0
---	---	---	---	---	---	---	---

A logical right shift of **three** places is performed on the binary number.

- (i) Complete the binary register to show its contents after this logical right shift.

--	--	--	--	--	--	--	--

[1]

- (ii) State **one** effect this logical shift has on the binary number.

.....

..... [1]

- (c) Give **two** reasons why a programmer may use hexadecimal to represent binary numbers.

1

.....

2

.....

[2]

- (d) Denary numbers can also be converted to hexadecimal.

Convert the denary number to hexadecimal.

301 [2]

Working space

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.....

Question 3:

Computers store data as binary. The binary number 10101110 is stored.

- (a) Convert the binary number to denary.

..... [1]

Working space

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.....
.....

- (b) Convert the binary number to hexadecimal.

..... [2]

Working space

.....
.....
.....

- (c) A logical left shift of **three** places is performed on the binary number.

- (i) Give the 8-bit binary number that would be stored after this logical left shift.

..... [1]

- (ii) Tick (✓) **one** box to show which statement is true about the impact the logical left binary shift would have on the binary number.

A The least significant bits are lost.

☐

B The most significant bits are lost.

☐

C The number has been divided by six.

☐

D The number stays the same.

☐

[1]

(d) Add the **two** 8-bit binary numbers 11101110 and 00110001 using binary addition.

Give your answer in binary. Show all your working.

.....

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..... [4]

(e) The denary number 301 needs to be stored.

Calculate the least number of bits that can be used to store the denary number 301.

..... [1]

Working space

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.....

.....

(f) The hexadecimal number A4D needs to be stored.

Calculate the least number of bits that can be used to store the hexadecimal number A4D.

..... [1]

Working space

.....

Question 4:

All data needs to be converted to binary data so that it can be processed by a computer.

(a) Explain why a computer can only process binary data.

.....
.....
.....
..... [2]

(b) The denary values 64, 101 and 242 are converted to 8-bit binary values.

Give the 8-bit binary value for each denary value.

64
101
242 [3]

Working space

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.....
.....

(c) The hexadecimal values 42 and CE are converted to binary.

Give the binary value for each hexadecimal value.

42
CE [4]

Working space

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.....
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.....

Question 5:

- (a) Denary values are converted to binary values to be processed by a computer.

Draw **one** line from each denary value to the correctly converted 8-bit binary value.

Denary	8-bit binary
	00100001
41	10100110
	00101001
174	10000110
	10101110
86	01010110

[3]

Working space

.....

.....

- (b) Binary values can also be converted to denary values.

Give the correct denary value for the 12-bit binary value 000101010111
Show all your working.

.....

.....

.....

.....

Denary value

[2]

Question 6:

Hexadecimal is used for Hypertext Markup Language (HTML) colour codes.

An HTML colour code is:

#2F15D6

Each pair of digits is stored as binary in an 8-bit register.

(a) Give the 8-bit binary value that would be stored for each pair of hexadecimal digits.

2F							
15							
D6							

[6]

Working space

.....

.....

.....

Question 7:

An aeroplane has a small display screen above each seat, to display the seat number.

(a) The seat number is a hexadecimal value. A 12-bit binary register is used to store the data to display each seat number.

Three seat numbers, 05A, 18C and 29F, are allocated to passengers.

Give the 12-bit binary value that would be stored for each hexadecimal seat number.

05A

18C

29F

[6]

Working space

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.....

(b) The display screen used above each seat is a Light-Emitting Diode (LED) display screen.

Give **two** reasons why an LED display screen has been used.

Reason 1

.....

Reason 2

.....

[2]

(c) **Two** of the registers store the values 010000001101 and 000001111110

Give the hexadecimal seat number that would be displayed on the screen for each of these binary values.

010000001101

000001111110

[4]

Working space

.....

.....

.....

.....

Question 8:

Benedict has a computer that is assigned an Internet Protocol (IP) address. The IP address is:

198.167.214.0

The IP address is represented as denary values.

(a) Convert the denary values 167 and 214 from the IP address to 8-bit binary.

167

--	--	--	--	--	--	--	--

214

--	--	--	--	--	--	--	--

Working space

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.....

.....

[2]

(b) Benedict's computer is also assigned a Media Access Control (MAC) address.

(i) Identify **one** similarity between an IP address and a MAC address.

.....
..... [1]

(ii) Identify **two** differences between an IP address and a MAC address.

Difference 1
.....
.....
Difference 2
.....
..... [2]

Question 9:

A denary value can be converted into hexadecimal and binary.

(a) Complete the table to show the hexadecimal and 8-bit binary values of the given denary values.

Denary	Hexadecimal	8-bit binary
49		
123		
200		

[6]

(b) Give **two** benefits, to users, of converting binary values to hexadecimal.

Benefit 1

.....

Benefit 2

.....

[2]

(c) Hexadecimal is used to represent Hypertext Markup Language (HTML) colour codes in computer science.

Identify **three** other ways that hexadecimal is used in computer science.

1

2

3

[3]

Question 10:

Greta has a computer that she uses for schoolwork and leisure.

(a) The computer has the Media Access Control (MAC) address:

00:A0:C9:14:C8:29

(i) Tick (✓) to show whether the MAC address is initially assigned to the computer by the network, the manufacturer or the user.

Tick (✓)

Network ☐

Manufacturer ☐

User ☐

[1]

- (ii) The values in the MAC address are hexadecimal values.

Convert the **three** given hexadecimal values into 8-bit binary.

14

A0

C9

[3]

Working space

.....
.....
.....

- (iii) Convert the **two** given hexadecimal values into denary.

29

C8

[2]

Question 11:

- (a) Give the **denary** value of each of the three 12-bit binary values.

(i) 000000001100

..... [1]

(ii) 000011000110

..... [1]

(iii) 010011000001

..... [1]

- (b) 12-bit binary values can also be represented as hexadecimal values.

Give the **hexadecimal** value of the 12-bit binary value.

000011101001

..... [3]

Question 12:

Pradeep is reading hexadecimal values for a project he is working on.

- (a) The first three hexadecimal values he reads are **15**, **102** and **A9**.

Give the **denary** values for the three hexadecimal values.

15

102

A9

[3]

Working space

.....

.....

.....

.....

- (b) Pradeep has two 8-bit binary values that he needs to convert to hexadecimal values for his project.

Give the **hexadecimal** values for the two 8-bit binary values.

01010000

00111101

[4]

Question 13:

Programmers can use denary and hexadecimal values. These values are stored in a computer system using binary.

- (a) Explain why binary is used to store data in a computer system.

.....

.....

.....

..... [2]

- (b) Complete the table to show how the denary value would be stored as binary in an 8-bit register.

Denary value	8-bit register
129	
56	

[2]

- (c) Complete the table to show how the hexadecimal value **3A9** would be stored as binary in a 12-bit register.

--	--	--	--	--	--	--	--	--	--	--	--

[3]

- (d) Identify **two** uses of hexadecimal values in computer science.

1

2

[2]

Question 14:

Hexadecimal is used for MAC addresses.

Part of a MAC address is given:

97 – 5C – E1

Each pair of digits is stored as binary in an 8-bit register.

- (a) Show what the binary register stores for each pair of the given digits.

97	<table border="1"><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								
5C	<table border="1"><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								
E1	<table border="1"><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								

[6]

Give **two** other examples where hexadecimal can be used.

Example 1

.....

Example 2

.....

[2]

Question 15:

Victoria is building a website for her cake design business.

- (a) She uses the hexadecimal colour code #D2E3F5 as the background colour for her website.

The colour code is stored in two 12-bit binary registers.

Show how the code would be stored in the registers.

D2E

--	--	--	--	--	--	--	--	--	--	--	--

3F5

--	--	--	--	--	--	--	--	--	--	--	--

[6]

Question 16:

- (a) **Six** binary or hexadecimal numbers and **six** denary conversions are given.

Draw a line to connect each binary or hexadecimal number to the correct denary conversion.

Binary or hexadecimal

Denary

01001011

75

4E

78

11011010

157

10011101

167

A7

25

19

218

[5]

- (b) Hexadecimal is often used by computer programmers to represent binary values.

Explain why computer programmers may choose to use hexadecimal.

.....

.....

.....

..... [2]

Question 17:

Computers use a character set to convert text into binary.

One character set that can be used is ASCII.

Each letter in ASCII can also be represented as a denary value.

- (a) The word BUS has the denary values:

B	U	S
66	85	83

Convert the denary values into 8-bit binary.

66

--	--	--	--	--	--	--	--

85

--	--	--	--	--	--	--	--

83

--	--	--	--	--	--	--	--

[3]

- (b) Each letter in ASCII can also be represented as a hexadecimal value.

The word KEY has the 8-bit binary values:

K	E	Y
01001011	01000101	01011001

- (i) Convert the three 8-bit binary values into hexadecimal.

01001011

01000101

01011001

[3]

(ii) Give **three** other uses of hexadecimal notation in computer science.

1

2

3

[3]

(iii) State **two** benefits of using hexadecimal notation to represent binary values.

Benefit 1

.....

Benefit 2

.....

[2]

Question 18:

The MAC address of a device is represented using hexadecimal.

A section of a MAC address is shown. Each pair of hexadecimal digits is stored using 8-bit binary.

(a) Complete the table to show the 8-bit binary equivalents for the section of MAC address. The first number has already been converted.

6A	FF	08	93
01101010			

[3]

(b) Explain why data is stored as binary in computers.

.....

.....

.....

.....

[2]

Question 19:

A washing machine has a small display screen built into it.

One use of the display screen is to show an error code when a problem has occurred with a washing cycle.

- (a) State whether the display screen is an **input**, **output** or **storage device**.

.....[1]

- (b) The display screen shows a hexadecimal error code:

E04

This error code means that the water will not empty out of the washing machine.

Convert this error code to binary.

--	--	--	--	--	--	--	--	--	--	--	--

[3]

- (c) State why hexadecimal is used to display the error code.

.....
.....
.....
.....[1]

Question 20:

A robot arm in a factory is programmed to move products.

The binary instructions to operate the robot arm are:

Operation	Binary Instruction			
UP	1	1	1	1
DOWN	0	0	0	1
LEFT	1	0	0	1
RIGHT	0	1	1	0
OPEN	1	1	0	0
CLOSE	0	0	1	1

The instructions are entered as hexadecimal values.

An operator enters the values:

9 1 C 3 F

Convert the values and write down the operation (e.g. RIGHT) carried out by the robot arm.

9

1

C

3

F

[5]

1.2 Text, sound and images:

Question 1:

A student has a sound file that is too large to be stored on their external secondary storage device. The student compresses the sound file to make the file size smaller.

The compression method used reduces the sample rate and the sample resolution of the sound file.

- (a) State what is meant by the sample rate and sample resolution.

Sample rate
.....
Sample resolution
..... [2]

- (b) Identify which type of compression has been used to compress the sound file.

.....
..... [1]

Question 2:

When keys are pressed on a keyboard, the text is converted to binary to be processed by the computer.

- (a) Describe how the text is converted to binary to be processed by the computer.

.....
.....
.....
.....
.....
..... [3]

- (b) Text that is input into a computer can be stored in a text file.

A text file can be compressed using lossless compression.

- (i) State what effect this has on the file size.

.....
..... [1]

(ii) Describe how lossless compression compresses the text file.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(iii) Give **two** reasons why the text file may have been compressed.

1

.....

2

.....

..... [2]

Question 3

1.3 Data storage and compression:

Question 1:

Jack has an MP3 file stored on his computer.

(a) (i) Tick (✓) to show which type of data is stored in an MP3 file.

	Tick (✓)
Video	☐
Sound	☐
Image	☐

[1]

(ii) Tick (✓) to show whether the MP3 file is a lossy compressed file or a lossless compressed file or **not** a compressed file.

	Tick (✓)
Lossy compressed file	☐
Lossless compressed file	☐
Not a compressed file	☐

[1]

Question 2:

An image is stored on a computer. The image is 16-bit colour and is 100 pixels high and 150 pixels wide.

Calculate the file size of the image in bytes. Show all your working.

.....

.....

.....

.....

.....

Answer bytes

[3]

1.1 Number systems:

Answer 1:

Question	Answer	Marks
1(a)	B	1
1(b)	One mark per each correct conversion 00110010 01100110 11011101	3
1(c)	One mark for full method of working e.g. conversion to binary then flipping and adding 1 One mark for correct answer 10110010	2
1(d)	One marks per each correct nibble One mark for correct working in binary (showing 4 correct carries) 1 1 1 1 0 0 1 1 0 0 1 1 0 1 1 0 0 0 0 1 ----- 10 0 1 0 1 0 0	3
1(e)	Two from: - The result of the calculation is greater than 255 // The value generated is larger than can be stored in the register - The result of the calculation would require more than 8 bits to be represented // A register has a predetermined number of bits and there are too many bits for it	2

Answer 2:

Question	Answer	Marks
2(a)	One mark per each correct character in the correct order: • 9 • 3 • 0 • D	4
2(b)(i)	• 00001111	1
2(b)(ii)	Any one from: • The value becomes incorrect/inaccurate as the right most bits are lost • It is divided by 8	1
2(c)	Any two from: • Easier/quicker to understand/read/write • Easier/quicker to debug • Less likely to make a mistake • Shorter representation // Takes up less screen space	2
2(d)	One mark for two correct characters, two marks for three correct characters in the correct order: • 1 • 2 • D	2

Answer 3:

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Question	Answer	Marks
1(a)	• 174	1
1(b)	• A • E	2
1(c)(i)	• 01110000	1
1(c)(ii)	• B	1
1(d)	One mark for each correct nibble One mark for correct carries (or other correct working method) One mark for identification of overflow error 1 1 • 1 0001 1111	4
1(e)	• 9	1
1(f)	• 12	1

Answer 4:

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Question	Answer	Marks
4(a)	• Computer consist of transistors / logic circuits/gates ... • ... that can only store/process data in two states / high-low / on-off / 1 and 0	2
4(b)	• 01000000 • 01100101 • 11110010	3
4(c)	• 0100 (1 mark) 0010 (1 mark) • 1100 (1 mark) 1110 (1 mark)	4

Answer 5:

Question	Answer	Marks														
1(a)	One mark for each correct line <table><thead><tr><th>Denary</th><th>8-bit binary</th></tr></thead><tbody><tr><td></td><td>00100001</td></tr><tr><td>41</td><td>10100110</td></tr><tr><td></td><td>00101001</td></tr><tr><td>174</td><td>10000110</td></tr><tr><td></td><td>10101110</td></tr><tr><td>86</td><td>01010110</td></tr></tbody></table>	Denary	8-bit binary		00100001	41	10100110		00101001	174	10000110		10101110	86	01010110	3
Denary	8-bit binary															
	00100001															
41	10100110															
	00101001															
174	10000110															
	10101110															
86	01010110															
1(b)	One mark for correct working, one mark for correct answer Working e.g. 256 + 64 + 16 + 4 + 2 + 1 Answer: 343	2														

Answer 6:

Question	Answer	Marks																											
2(a)	Two marks each correct conversion (one mark for the first four bits, one mark for the second four bits) <table><tr><td>2F</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>15</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>D6</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td></tr></table>	2F	0	0	1	0	1	1	1	1	15	0	0	0	1	0	1	0	1	D6	1	1	0	1	0	1	1	0	6
2F	0	0	1	0	1	1	1	1																					
15	0	0	0	1	0	1	0	1																					
D6	1	1	0	1	0	1	1	0																					

Answer 7:

Question	Answer	Marks
2(a)	Per each binary conversion, one mark for 2 correct character conversions, two marks for three correct character conversions 0000 0101 1010 0001 1000 1100 0010 1001 1111	6
2(b)	Any two from: - It has a bright screen/colours - It does not consume much power - It runs at a cool temperature - Longevity - Can operate in cold conditions - Can be small/compact/thin	2
2(c)	Per each hex conversion, one mark for 2 correct character conversions, two marks for three correct character conversions 40D 07E	4

Answer 8:

Question	Answer	Marks																
1(a)	One mark per each correct register. <table border="1"><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td></tr></table>	1	0	1	0	0	1	1	1	1	1	0	1	0	1	1	0	2
1	0	1	0	0	1	1	1											
1	1	0	1	0	1	1	0											
1(b)(i)	Any one from: - Both addresses can be used to identify a computer/device - Both are unique - Both can be represented as hexadecimal - Both addresses do not change if IP address is static	1																
1(b)(ii)	Any two from: - An IP address is assigned by the network/router/ISP, A MAC address is assigned by the manufacturer - An IP address can be changed (if dynamic), MAC address cannot be changed - IP address has 4/8 groups of values, MAC address has 6 groups/pairs of values - IP address is 32-bit/128-bit, MAC address is 48-bit - IP address does not contain serial number/manufacture number, MAC address does - IP(v4) address is denary and MAC address is hexadecimal	2																

Answer 9:

Question	Answer	Marks												
1(a)	One mark per each correct binary value. One mark per each correct hex value. <table border="1"> <thead> <tr> <th>Denary</th><th>Hexadecimal</th><th>8-bit binary</th></tr> </thead> <tbody> <tr> <td>49</td><td>31</td><td>00110001</td></tr> <tr> <td>123</td><td>7B</td><td>01111011</td></tr> <tr> <td>200</td><td>C8</td><td>11001000</td></tr> </tbody> </table>	Denary	Hexadecimal	8-bit binary	49	31	00110001	123	7B	01111011	200	C8	11001000	6
Denary	Hexadecimal	8-bit binary												
49	31	00110001												
123	7B	01111011												
200	C8	11001000												
1(b)	Any two from: - Easier/quicker to read/write/understand - Easier/quicker to identify errors/debug - Takes up less screen/display space - Less chance of making an error	2												
1(c)	Any three from: - MAC address - URL - Assembly language - Error codes // error messages - IP addresses - Locations in memory - Memory dumps	3												

Answer 10:

Question	Answer	Marks
1(a)(i)	– manufacturer	1
1(a)(ii)	One mark per each correct binary value. – 00010100 – 10100000 – 11001001	3
1a(iii)	One mark per each correct denary value. – 41 – 200	2
1(b)(i)	– Magnetic – Solid state – Optical	3
1(b)(ii)	Any four from: – It has platters – Platters/disk divided into tracks – Platter/disk is spun – Has a read/write arm that moves across storage media – Read/writes data using electromagnets – Uses magnetic fields to control magnetic dots of data – Magnetic field determines binary value	4

Answer 11:

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Question	Answer	Marks
7(a)(i)	– 12 (ignore leading zeros)	1
7(a)(ii)	– 198 (ignore leading zeros)	1
7(a)(iii)	– 1217	1
7(b)	One mark per each correct hex value in correct order – 0E9	3

Answer 12:

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Question	Answer	Marks
1(a)	– 21 – 258 – 169	3
1(b)	1 mark for each correct hex value – 50 – 3D	4

Answer 13:

Question	Answer	Mark						
5(a)	Any two from: • Computer consist of transistors / logic circuits • ... that can only store/process data in two states / as high-low / on-off / 1 and 0	2						
5(b)	1 mark per each correct 8-bit binary value: <table><tr><th>Denary Value</th><th>8-bit binary register</th></tr><tr><td>129</td><td>10000001</td></tr><tr><td>56</td><td>00111000</td></tr></table>	Denary Value	8-bit binary register	129	10000001	56	00111000	2
Denary Value	8-bit binary register							
129	10000001							
56	00111000							

Question	Answer	Mark												
5(c)	1 mark per each correct conversion: <table><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr></table> 1 mark 1 mark 1 mark	0	0	1	1	1	0	1	0	1	0	0	1	3
0	0	1	1	1	0	1	0	1	0	0	1			
5(d)	Any two from: • Represent colours in HTML // HTML colour codes • MAC address • Assembly Language • Error messages • IP address • ASCII values • URL • Memory dump • Memory locations	2												

Answer 14:

Question	Answer								Marks									
1(a)	97	<table><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr></table> 1 mark 1 mark								1	0	0	1	0	1	1	1	6
	1	0	0	1	0	1	1	1										
	5C	<table><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table> 1 mark 1 mark								0	1	0	1	1	1	0	0	
0	1	0	1	1	1	0	0											
E1	<table><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table> 1 mark 1 mark								1	1	1	0	0	0	0	1		
1	1	1	0	0	0	0	1											

Question	Answer	Marks
1(c)	Two from e.g.: • Colour codes // Colour in HTML / CSS • Error messages • Locations in memory • Memory dump // debugging • IP address • ASCII // Unicode • Assembly language • URL	2

Answer 15:

Question	Answer	Marks
1(a)	The first row of binary is 1 1 0 1 0 0 1 0 1 1 1 0. Brackets below indicate 1 mark for each group of 4 bits. The second row of binary is 0 0 1 1 1 1 1 1 0 1 0 1. Brackets below indicate 1 mark for each group of 4 bits.	6

Answer 16:

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Question	Answer	Marks																					
2(a)	1 mark for each correct line (to a maximum of 5) <table><thead><tr><th>Binary or hexadecimal</th><th></th><th>Denary</th></tr></thead><tbody><tr><td>01001011</td><td>—</td><td>75</td></tr><tr><td>4E</td><td>—</td><td>78</td></tr><tr><td>11011010</td><td>—</td><td>157</td></tr><tr><td>10011101</td><td>—</td><td>167</td></tr><tr><td>A7</td><td>—</td><td>25</td></tr><tr><td>19</td><td>—</td><td>218</td></tr></tbody></table>	Binary or hexadecimal		Denary	01001011	—	75	4E	—	78	11011010	—	157	10011101	—	167	A7	—	25	19	—	218	5
Binary or hexadecimal		Denary																					
01001011	—	75																					
4E	—	78																					
11011010	—	157																					
10011101	—	167																					
A7	—	25																					
19	—	218																					
2(b)	Two from: ∞ It makes the values easier to read/write/understand/debug ∞ It is a shorter way to represent the values	2																					

Answer 17:

Question	Answer	Marks																								
1(a)	1 mark for each correct 8-bit binary number 66 <table border="1"><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr></table> 85 <table border="1"><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr></table> 83 <table border="1"><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>	0	1	0	0	0	0	1	0	0	1	0	1	0	1	0	1	0	1	0	1	0	0	1	1	3
0	1	0	0	0	0	1	0																			
0	1	0	1	0	1	0	1																			
0	1	0	1	0	0	1	1																			
1(b)(i)	1 mark for each correct hexadecimal number 4B 45 59	3																								
1(b)(ii)	Three from: ∞ (HTML) colour codes ∞ Error messages ∞ MAC addresses ∞ IP addresses ∞ Assembly language ∞ Memory dump ∞ Locations in memory	3																								
1(b)(iii)	Two from: ∞ Easier to read/write/understand (for humans) ∞ Easier to remember (for humans) ∞ Short way to represent binary // Uses less screen/display space ∞ Fewer errors made (in data transcription) ∞ Easier to debug (for humans)	2																								

Answer 18:

Question	Answer				Marks
4(a)	1 mark for each correct conversion				3
	01101010	11111111	00001000	10010011	
4(b)	α Computers use switches / logic gates ∞ Only uses 2 states / On or Off / 1 or 0				2

Answer 19:

Question	Answer	Marks												
1(a)	Output	1												
1(b)	1 mark for each correct conversion E 0 4 <table><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr></table>	1	1	1	0	0	0	0	0	0	1	0	0	3
1	1	1	0	0	0	0	0	0	1	0	0			
1(c)	Any one from: - Hexadecimal codes can fit in a smaller display rather than a full text based message - Smaller amount of memory needed to store the hex error messages than text based	1												

Answer 20:

Question	Answer	Marks
1	1 mark per correct instruction: 9 – LEFT 1 – DOWN C – OPEN 3 – CLOSE F – UP	5

1.2 Text, sound and images:

Answer 1:

Question	Answer	Marks
2(a)	One mark for each correct definition: - The sample rate is the number of samples taken in a second/per time unit - The sample resolution is the number of bits per sample	2
2(b)	Lossy compression	1

Answer 2:

Question	Answer	Marks
3(a)	Any three from: - A character set is used ... such as Unicode/ASCII - Each character has a unique binary value	3
3(b)(i)	It reduces the file size	1
3(b)(ii)	Any four from: - A compression algorithm is used ... such as RLE/run length encoding Repeating words/characters/phrases are identified // <u>Patterns</u> are identified ... and indexed ... with number of occurrences ... with their position	4
3(b)(iii)	Any two from: e.g. - To save storage space - To make it quicker to transmit - To make it small enough to attach to an email - To reduce the bandwidth needed to transmit	2

1.3 Data storage and compression:

Answer 1:

Question	Answer	Marks
1(a)(i)	Sound	1
1(a)(ii)	Lossy compressed file	1

Answer 2:

Question	Answer	Marks
5	Two marks for two correct stages of working, one mark for correct final answer 100×150 $15\,000 \times 16 // 15\,000 \times 2$ $240\,000 / 8$ 30 000 bytes	3